

Research



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Population ecology

Partitioning colony size variation into growth and partial mortality

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Body size is a trait that broadly influences the demography and ecology of organisms. In unitary organisms, body size tends to increase with age. In modular organisms, body size can either increase or decrease with age, with size changes being the net difference between modules added through growth and modules lost through partial mortality. Rates of colony extension are independent of body size, but net growth is allometric, suggesting a significant role of size-dependent mortality. In this study, we develop a generalizable model of partitioned growth and partial mortality and apply it to data from 11 species of reef-building coral. We show that corals generally grow at constant radial increments that are size independent, and that partial mortality acts more strongly on small colonies. We also show a clear life-history trade-off between growth and partial mortality that is governed by growth form. This decomposition of net growth can provide mechanistic insights into the relative demographic effects of the intrinsic factors (e.g. acquisition of food and life-history strategy), which tend to affect growth, and extrinsic factors (e.g. physical damage, and predation), which tend to affect mortality.

1. Introduction

Body size is a fundamental organismal trait [1], which determines a range of physiological, demographic and ecological possibilities and outcomes [2]. For example: large-bodied organisms have lower per-mass metabolic rates than small-bodied organisms [3]; population densities of large species are expected to be lower than those of smaller ones because of their higher total metabolic demands [4]; and declines in average body size can foreshadow population declines [5]. An organism's ability to increase body size is its growth. For unitary organisms, growth is the only rate required to determine body size because they generally do not shrink, although losses in biomass are possible from decreased body condition. However, modular organisms can shrink, because modules can die without killing the whole organism (hereafter termed 'partial mortality'). For example, herbivory or mechanical damage to such organisms can more than outpace growth and accretion rates [6,7]. For this reason, a mechanistic understanding of growth in modular organisms requires that the two processes of growth and partial mortality be estimated separately. Here, we use scleractinian reef-building coral colonies to build a model that separates the intrinsic capacity to increase in size from the processes that act to reduce it.

We use scleractinian corals because the processes that add or reduce body size are well understood, with the addition of modules by asexual reproduction counteracted by the loss of modules through partial mortality [6]. If we were only interested in the net change in body size, the effects of these counteracting

Table 1. Regression intercept estimates for upper 95% quantiles of added radius for the 11 species of reef-building coral, ranked from highest to lowest. Estimates are potential radial growth, r_s , for species in metres per year.

species, s	growth form	estimated r_s (m yr ⁻¹)	s.e.	t-value	Pr(> t)
<i>Acropora cytherea</i>	tabular	0.096	0.018	5.353	<0.001
<i>Acropora hyacinthus</i>	tabular	0.070	0.006	12.321	<0.001
<i>Acropora robusta</i>	arborescent	0.049	0.010	4.874	<0.001
<i>Acropora intermedia</i>	arborescent	0.044	0.011	4.151	<0.001
<i>Acropora spathulata</i>	corymbose	0.033	0.004	7.615	<0.001
<i>Acropora nasuta</i>	corymbose	0.024	0.002	10.576	<0.001
<i>Acropora millepora</i>	corymbose	0.024	0.003	7.323	<0.001
<i>Acropora cf. digitifera</i>	digitate	0.019	0.004	4.436	<0.001
<i>Acropora humilis</i>	digitate	0.018	0.002	7.270	<0.001
<i>Goniastrea retiformis</i>	massive	0.014	0.002	6.671	<0.001
<i>Goniastrea pectinata</i>	massive	0.014	0.004	3.376	<0.001

processes could be considered only in the aggregate. For instance, demographic modelling is typically concerned with net changes in colony size through time and less-so with how these changes come about [8,9]. For corals, net growth tends to be allometric [10]. However, annual rates of linear extension in the absence of significant partial mortality (e.g. increasing branch length or colony radius) tend to be constant and independent of colony size [11,12]. We hypothesize that these apparently conflicting findings can be resolved by separating the effects of partial mortality from the ability to grow.

To do so, we developed a generalizable approach that separates change in colony size into its potential growth and partial mortality components. Using annual growth measurements across 11 reef-building coral species, we then tested the hypotheses that (1) linear extension is constant but (2) partial mortality is size dependent. Finally, we tested for a relationship between potential growth and partial mortality which may represent a fundamental life-history trade-off for reef corals.

2. Material and methods

We estimated yearly changes in planar area for 11 species of scleractinian corals. We tagged 30 colonies of each species, which were distributed along a 500 by 10 m band of reef crest at Trimodal Reef, between South and Palfrey islands (14.6998398 S, 145.4486748 E), Lizard Island, Australia. Each tagged colony was photographed from above with a two-dimensional scale plate placed level with the surface of the colony (electronic supplementary material, figure S1). The angle of the camera was horizontal, and the distance from the colony was such that the entire colony was visible in the photograph. Tagged colonies were photographed yearly in November from 2009 to 2013. The images were corrected for barrel distortion, and the scale and outline of each colony were digitized in ImageJ to estimate colony size (electronic supplementary material, figure S1). Every year, dead or missing colonies were replaced in order to maintain approximately 30 colonies per species. We did this to ensure that size-selective mortality would not progressively reduce the range of colony sizes available to characterize size-dependent growth. To minimize the effect of observation error, colonies were photographed two or three times independently every year. All photographs were digitized twice independently and estimates of area were averaged.

To test the constant rate of linear growth hypothesis, we first calculated, for each colony, the radius of a hypothetical circular colony with the same planar area as the corresponding real colony. We then calculated the yearly change in radii for each individual colony that survived over a given time interval and, for a given species, assumed that colonies with the greatest positive radial change per year were those suffering the least partial mortality (i.e. colonies can grow at least at this rate). We used quantile regression models to capture the extreme upper bound (95% quantile) of relationships between annual added radius and the planar area of a colony for each species separately. Because some colonies were measured over multiple years, we first used linear quantile mixed effect models with colony ID as a random intercept, using 'lqmm' [13] in the statistical software R [14]. Variance of the random effect was consistently much lower than differences among model estimates—suggesting that colony growth of individuals could be treated independently from year to year—and so we removed the random effect and used basic quantile regression instead, using the 'rq' function [15]. We tested if the slope of the upper quantile as a function of colony planar area was statistically indistinguishable from zero (i.e. size independent). Given 11 species comparisons, we applied a Bonferroni correction to the p -value. For a colony of planar area x at time t , yearly potential increase in colony area (growth) g from radial growth r_s for each species s was given by

$$g(x, r_s) = \pi \left(\sqrt{\frac{x}{\pi}} + r_s \right)^2, \quad (2.1)$$

where the square-root term is the initial radius of a circular colony of planar area x . To test if our assumption of colony circularity affected the results, we calculated the circularity of colonies as the ratio of colony digitized outline lengths and the perimeter of a circle with the same planar area. (Thus, a perfectly circular colony would have a ratio of 1.) Colony residuals of the quantile regression models for each species (table 1) were plotted against colony circularity values, and Spearman's correlations were calculated to quantify the strength of associations.

Partial mortality operates over the range between maximum potential growth (i.e. no partial mortality) and whole-colony death (i.e. partial mortality that is equal to colony size). We estimated partial mortality, p , as the proportion of area lost across years following potential growth, which is 1 minus the area remaining, given by

$$p(x, y, r_s) = 1 - \frac{y}{g(x, r_s)}, \quad (2.2)$$

Table 2. Regression estimates for linear models of logit partial mortality as a function of colony size for the 11 species of reef-building coral. σ_s is residual standard deviation for species.

species, <i>s</i>	parameter	estimate	s.e.	t-value	Pr(> t)	σ_s
<i>Acropora hyacinthus</i>	<i>c</i> (intercept)	−0.963	0.3	−3.213	0.002	1.397
	<i>m</i> (slope)	−1.177	0.311	−3.786	<0.001	
<i>Acropora cytherea</i>	<i>c</i>	−0.276	0.205	−1.35	0.181	0.851
	<i>m</i>	−1.054	0.222	−4.753	<0.001	
<i>Acropora intermedia</i>	<i>c</i>	−0.468	0.408	−1.146	0.258	1.177
	<i>m</i>	−1.042	0.283	−3.68	0.001	
<i>Acropora robusta</i>	<i>c</i>	−0.518	0.254	−2.037	0.044	0.899
	<i>m</i>	−1.172	0.216	−5.433	<0.001	
<i>Acropora cf. digitifera</i>	<i>c</i>	−1.646	0.252	−6.541	<0.001	0.656
	<i>m</i>	−0.939	0.174	−5.402	<0.001	
<i>Acropora humilis</i>	<i>c</i>	−1.394	0.293	−4.749	<0.001	0.589
	<i>m</i>	−1.134	0.172	−6.596	<0.001	
<i>Acropora spathulata</i>	<i>c</i>	−1.847	0.596	−3.099	0.003	1.213
	<i>m</i>	−1.236	0.422	−2.926	0.004	
<i>Acropora nasuta</i>	<i>c</i>	−1.347	0.631	−2.135	0.035	1.268
	<i>m</i>	−0.868	0.441	−1.966	0.052 ^a	
<i>Acropora millepora</i>	<i>c</i>	−0.837	0.836	−1.001	0.320	1.410
	<i>m</i>	−0.512	0.509	−1.005	0.319 ^a	
<i>Goniastrea retiformis</i>	<i>c</i>	−2.927	0.313	−9.362	<0.001	0.685
	<i>m</i>	−1.537	0.172	−8.925	<0.001	
<i>Goniastrea pectinata</i>	<i>c</i>	−1.158	0.435	−2.663	0.009	0.747
	<i>m</i>	−0.869	0.198	−4.394	<0.001	

^aSpecies without a significant slope with a Bonferroni correction α of 0.0045.

where y is the area at time $t + 1$. To test if colony size affected the proportional partial mortality, we used linear mixed effects models for species, where partial mortality was logit transformed and colony ID was the random intercept, using 'lmer' [16]. As for quantile regressions, the random effect accounting for the same individuals measured in multiple years explained little variation and was removed.

Finally, we calculated the combined annual growth probability f for each study species s by combining yearly potential growth (equation (2.1)) and partial mortality (equation (2.2)) probabilities based on colony size. We used the normal density function to model the probability of annual growth to size y given a starting size x and, for species s , the yearly added radius r_s and the slope m_s , intercept c_s and error σ_s parameters for partial mortality. The growth component was logit transformed before calculating the probability of attaining size y given partial mortality, and the result back-transformed using the inverse logit as follows

$$f(x, y, r_s, m_s, c_s, \sigma_s) = \text{logit}^{-1} \left(\frac{1}{\sqrt{2\pi}} e^{-([\text{logit}(p(x, y, r_s)) - m_s x + c_s]^2 / 2\sigma_s^2)} \right). \quad (2.3)$$

3. Results

The maximum potential growth (measured as radial extension) of coral colonies did not change significantly with colony size (electronic supplementary material, figure S2 and table S1). Given general support for constant radial

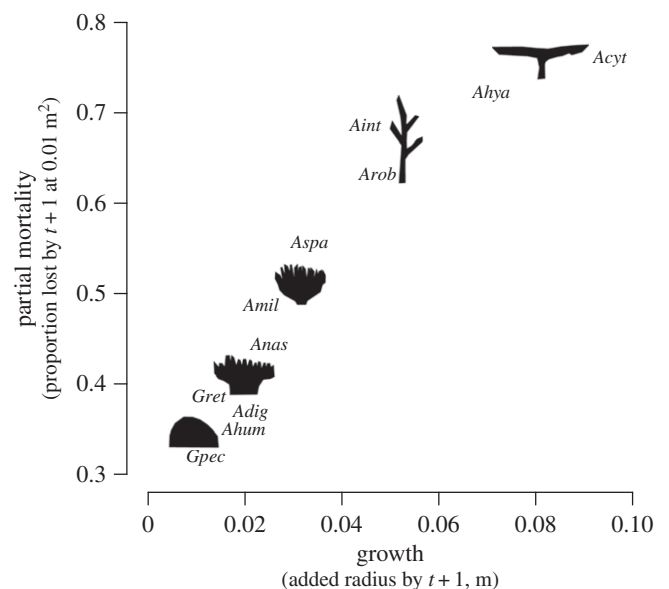


Figure 1. The association between yearly extension (added radius) and proportion partial mortality for the 11 study species. Points are shortened versions of species names (see tables). The proportion partial mortality was estimated for a colony size of 0.01 m² (equivalent to a 10 by 10 cm colony) for each species. Silhouettes of species growth forms are superimposed.

growth across species, we present intercept-only model estimates in table 1, which gives the rank order of potential yearly extension. Potential growth rate was greatest in the

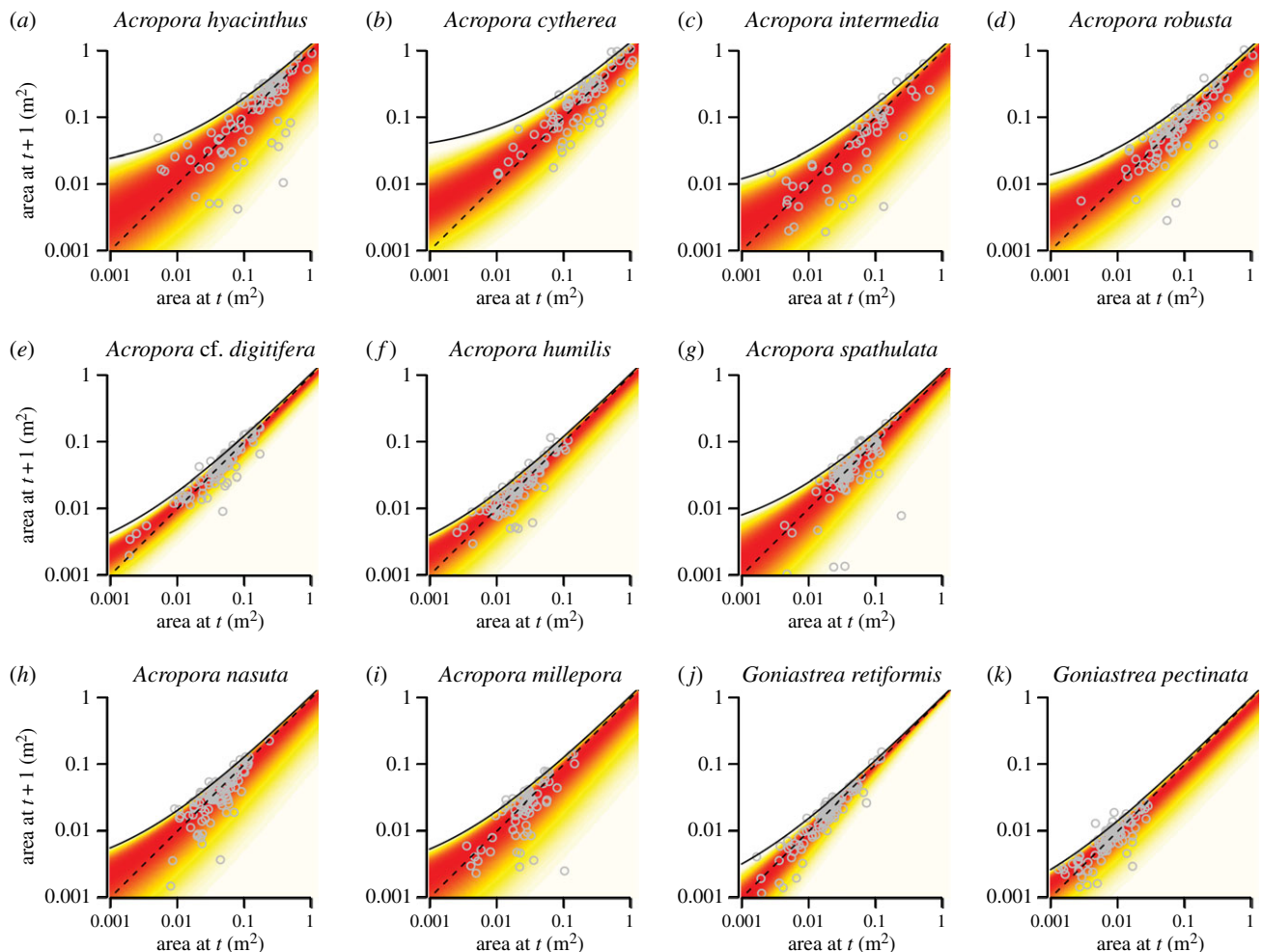


Figure 2. Planar area of colonies at $t+1$ as a function of the area at time t for the 11 study species. Points show the field data. The dashed line is the stasis line. The solid line is the maximum potential growth (i.e. added radial growth and no partial mortality). Shading represents the estimated probability of colony size at $t+1$ given size at t (red is the band of highest probability).

tabular species (*Acropora hyacinthus* and *Acropora cytherea*), which were estimated to add between 7 and 10 cm radially each year in the absence of partial mortality. Potential growth was lowest for the massive species (*Goniastrea retiformis* and *G. pectinata*) at between 1 and 2 cm yr^{-1} . No strong or consistent associations were found between circularity and growth model residuals (Spearman's ρ ranging between -0.191 and 0.368), suggesting that departure from circularity had a small effect on the extension rate estimates (electronic supplementary material, figure S3).

Generally, yearly partial mortality decreased with colony size (table 2; electronic supplementary material, figure S4), although the relationships were not significant for *Acropora nasuta* and *Acropora millepora*. The life-history trade-off between growth and partial mortality is presented in figure 1. The combined growth model (equation (2.3)) based on growth and partial mortality parameter estimates from tables 1 and 2 is presented in figure 2.

4. Discussion

In this study, we developed a growth model that separates maximum potential growth from partial mortality. In doing so, we show that reef corals' growth is generally consistent with constant annual radial increments, as previously suggested [11,17]. Change in the perimeter to area ratio as colonies grow causes a curvilinear change in maximum potential

growth (figure 2, solid upper curves), with the curve particularly evident for small sizes. This curve implies that size increase (in the absence of partial mortality) deviates from allometric growth. However, net growth—which includes both growth and partial mortality—tends to be allometric [10]. This inconsistency occurs because partial mortality is size dependent and acts to suppress the proportionally large size gains when colonies are small (electronic supplementary material, figure S3), which suppresses the upwards inflection of potential growth among smaller colonies.

Growth and partial mortality trade-off strongly, where faster-growing species have higher rates of partial mortality than slower-growing species (figure 1). The position of species along the trade-off dimension was determined by growth form (shown as silhouettes in figure 1), showing a clear pattern from slow-growing mechanically robust to fast-growing delicate colony shapes [18,19]. The trade-off also helps explain similarities in net growth when there are substantial differences in radial growth (figure 2). The consistency in patterns in potential growth and partial mortality across the 11 species in this study suggests that our findings are robust and should generalize to other species based on morphological traits.

One implication of this result is that estimates of growth based on estimates from large colonies will underestimate growth of recruits and small colonies, and hence the potential reef recovery rates following a disturbance. As a consequence, growth in very small colonies can be disproportionately fast

relative to their size, especially in the absence of partial mortality. The potential to increase colony area when small by over an order of magnitude in 1 year (figure 2, solid upper curve) is the mechanism that underlies the efficacy of micro-fragmenting (the chopping up of large colonies to increase overall areal growth rate) for reef restoration as it allows rapid generation of a large number of colonies for outplanting [20].

Separating growth and partial mortality is important for modelling the dynamics of modular organisms, because the two components are influenced by different biological, ecological and environmental processes. Growth is influenced by access to resources and life-history strategy—that is, the pattern of allocation of energy among reproduction, growth, maintenance and repair [21]. In turn, each of these processes is influenced by the environment and interactions with other organisms. On the other hand, partial mortality of colonies is more influenced by external processes, such as competition, predation and damage, for example, from waves [6]. Separating the intrinsic and extrinsic processes that lead to net growth will also allow improved, mechanistic predictions of the ecological and evolutionary consequences of environmental changes. For example, reduced seawater pH will affect carbonate availability for building skeletons, and therefore might result in a trade-off between growth rate and skeletal density. On the one end of this trade-off, maintaining a given growth rate at the cost of lower

skeletal density might increase the potential for partial and whole-colony mortality. At the other end of this trade-off, maintaining skeletal density might result in slower growth and the consequent reduction in reproductive output [22] and competitive capacity [23]. Modelling mechanistic trade-offs such these at the population level to inform expectations under future environmental change requires the separation of growth into its constituent parts.

Data accessibility. Data were previously published as supplemental data at [10] (<https://doi.org/10.1098/rspb.2017.0053>). The code and data are available at https://github.com/jmadin/coral_growth.

Authors' contributions. All authors designed the study. J.S.M., A.H.B., S.R.C. and M.A.D. collected the data. J.S.M. analysed the data and wrote the first draft. All authors contributed to revisions, approved the final version of the manuscript and agree to be held accountable for the content therein.

Competing interests. We declare we have no competing interests.

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